

capabilities and high brightness light emitting diode (LED) for visual indication to the user.

What sets it apart

The key differentiation of SafeDrive is automatic crash detection and notification mechanism.

The other products currently available in the market are limited to providing location-tracking features. SafeDrive adds a new spin to this by adding an in-built crash sensing and notification mechanism. The team claims that crash sensing is the key technology that differentiates this product from the others.

In conventional systems, false triggers for small impacts, including sudden braking and impacts from traversing large gutters on the road, affect the integrity of the solution in the long run.

If a system such as this is allowed to cause a false accident trigger, the ensuing situation can wreak havoc for the driver and entire support system. It will cause the solution to degrade and, consequently, affect its ability to be put to use in an effective manner. "We accomplish a stable, error-free and dependable accident detection system implemented through proprietary algorithms that contextually analyse and combine all the information obtained. This includes position, velocity, impact direction and duration, make and model of the vehicle, and so on," explains Pillai.

Current competition

In most developed and developing nations, including China, modern technologies are employed for fast detection and co-ordination of emergency rescue efforts.

In the USA, General Motors developed and commissioned OnStar system in their cars, that provides automatic accident detection and emergency SOS button, over 15 years ago. Currently, this system provides many services, including instant on-call support from a central office and dispatching emergency assistance to

What Happens in Case of an Accident?

SafeDrive scenario	A typical accident scene
1. As soon as the in-built sensors detect an accident, a quick notification is automatically sent with information relating to location and accident severity, among others, to emergency assistance centre and local emergency service providers.	1. The most common mode of notification is by kind-hearted bystanders who would call authorities after they see an accident.
2. Easy access is provided to emergency contacts, and their support centre will ensure proper care and attention to people in distress.	2. With the hustle and chaos in case of an accident, many things get overlooked easily. For example, in most cases, the emergency contacts, identity and medical history of the victim may not be known immediately.
3. Live support staff trained in dealing with emergency and crisis management will co-ordinate and streamline relief operations with local and government authorities and with the family of the crash victim.	3. In most cases, the first help offered to an accident victim is from mostly unprepared and confused, but helpful bystanders. This is not a very effective system.
4. With access to secure medical information provided from the user's profile, emergency responders can respond to the crisis faster and more efficiently.	4. The critical time for providing medical help is wasted, which adversely affects the chances of survival of the victim.
Statistics show that... Third-party studies show that about 50 per cent of lives can be saved if they are given medical attention within the first hour following an accident, often called the 'golden hour.'	

Solving challenges: Accident detection

The ability to detect acceleration forces of up to 100g (where g is the unit of acceleration; 1g=9.8 metres per second) was necessary for Raksha SafeDrive to accurately determine the impact and severity of the accident. Other aspects involved filtering methods to accurately identify false positives. The team did not want the system to trigger an accident for a false event, like if a device gets dropped on the floor. Identifying such false cases from an actual accident was also given due consideration.

the accident location, among others. Over the past several years, most other car manufacturers have followed their lead and have provided similar safety systems for their cars sold in developed countries like the USA, the UK and others.

However, in India, we are yet to see such technologies being used and made available for public consumption. Lack of such modern technologies for emergency response is a major problem to be dealt with in order to reduce high casualties in India from road accidents.

SafeDrive was available for pre-order in November at a price of ₹ 6999 from the company's website *Raksha.me*. Competing international products like Splitsecnd cost around ₹ 12,000 to buy, plus a monthly expense of ₹ 900

to subscribe to the emergency service, apart from routing service from a centre placed out of India.

Prasad adds, "We have taken two years, with the effort of a seven-member team, to bring this product into fruition."

What we can expect in the future

Prasad and his team are working on patenting some features, which they cannot disclose at this time due to patent filing requirements. From my interaction with the team, I am sure that we can expect some interesting improvements in the feature set that comes with the next generation of this product. ●

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